

Week 4 Day 2 — Supervised Learning: Preprocessing & Model Evaluation

1. Objective

The objective of this task was to build the first supervised machine learning models for the Adult Census Income dataset while using a reproducible preprocessing pipeline. The models were trained to predict whether an individual earns more than **\$50,000 per year**, with **precision** remaining the primary evaluation metric because the business objective is to reduce unnecessary marketing outreach.

2. Preprocessing Choices

The Adult dataset contains both **numeric** and **categorical** features, so separate preprocessing pipelines were created using a `ColumnTransformer`.

- **Numeric Features**

The following steps were applied:

- **Median Imputation** using `SimpleImputer(strategy="median")`
- **Standardization** using `StandardScaler`

Median imputation was selected because it is more robust to extreme values than mean imputation. This is especially important because **capital-gain** and **capital-loss** are highly skewed with many outliers. Standardization was applied so that numeric features have a similar scale, allowing models such as Logistic Regression to train more effectively.

- **Categorical Features**

The following steps were applied:

- **Most-Frequent Imputation** using `SimpleImputer(strategy="most_frequent")`
- **One-Hot Encoding** using `OneHotEncoder(handle_unknown="ignore")`

Most-frequent imputation preserves the existing categories without introducing artificial values. One-Hot Encoding was chosen because the categorical variables have no natural ordering, making it more appropriate than ordinal encoding. Setting `handle_unknown="ignore"` allows the model to safely process unseen categories during testing.

Alternatives Considered

Several alternatives were considered but were not used:

- **Mean imputation** for numeric features (more sensitive to outliers)
- **Constant-value ("Missing") imputation** for categorical features
- **Target Encoding**, which may introduce data leakage if not implemented carefully
- **KNN Imputation**, which is computationally more expensive and offers limited benefit for this dataset

3. Model Comparison

Both supervised models were trained using the same preprocessing pipeline and evaluated on the untouched hold-out test set.

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Majority Baseline	0.761	0.000	0.000	0.000	0.500	0.239
Rule-Based Baseline	0.779	0.608	0.215	0.318	0.586	0.319
Logistic Regression	0.850	0.728	0.593	0.654	0.902	0.756
Decision Tree	0.814	0.615	0.599	0.607	0.770	0.495

Both machine learning models significantly outperformed the Day 1 baselines. Logistic Regression achieved the highest performance across almost every evaluation metric, especially the project's primary metric, **precision**. The Decision Tree achieved a similar recall but lower precision and overall predictive performance.

4. Model Interpretation

- **Logistic Regression**

The largest positive coefficients were associated with features such as **higher education**, **married-civ-spouse**, **capital gain**, and **executive or professional occupations**, indicating that these characteristics increase the likelihood of earning more than \$50,000 annually.

Conversely, negative coefficients were associated with features such as **never married**, **other-service occupations**, **own-child relationship**, and several lower-income occupations, suggesting a lower probability of belonging to the high-income class.

- **Decision Tree**

The first decision made by the tree was based on **marital status (Married-civ-spouse)**, followed by **capital gain** and **education level**, indicating that these are among the most informative features for predicting income.

However, the tree achieved **97.5% training accuracy** but only **81.4% testing accuracy**, indicating substantial overfitting. Although the decision rules appear reasonable, the model memorized patterns in the training data that did not generalize well to unseen examples.

5. Model Selection & Future Work

Based on the evaluation results, **Logistic Regression** will be carried forward for further development. It produced the best overall performance while remaining easy to interpret through its feature coefficients. Its higher precision also aligns with the business objective of minimizing unnecessary marketing outreach.

The preprocessing pipeline developed today will be reused in future experiments to ensure consistency and prevent data leakage. Future work will include tuning Logistic Regression regularization, pruning the Decision Tree to reduce overfitting, exploring transformations for the highly skewed **capital-gain** and **capital-loss** features, and further evaluating whether both **education** and **education-num** should be retained.